# DSA06 DATA HANDLING AND VISUALIZATION
# TABLEAU INTRODUCTION ACTIVITY — SLOT C

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## T1. Connect and Check

Data source: C_22.csv. The dataset contains 48 rows and 6 columns.

Fields: Month (date), Region (dimension), Sales (measure, ₹ lakh), Profit (measure, ₹ lakh), Customers (measure/count), and Discount_pct (measure, percentage).

Month was parsed as a date. Region and Month are dimensions; Sales, Profit, Customers, and Discount_pct are measures. The file contains January–December 2025 for four regions.

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**T1. Connect and check data**

Tableau-style worksheet • C_22.csv

| Month | Region | Sales | Profit | Customers | Discount_pct |
|---|---|---|---|---|---|
| 2025-01-01 | Andhra Pradesh | 42.1 | 2.1 | 455 | 9.2 |
| 2025-01-01 | Karnataka | 57.2 | -3.4 | 784 | 20.0 |
| 2025-01-01 | Kerala | 89.4 | 9.6 | 1019 | 3.4 |
| 2025-01-01 | Maharashtra | 76.5 | 2.7 | 831 | 18.4 |
| 2025-02-01 | Andhra Pradesh | 37.8 | -0.4 | 399 | 9.1 |
| 2025-02-01 | Karnataka | 63.8 | 3.2 | 772 | 13.6 |
| 2025-02-01 | Kerala | 95.8 | 7.5 | 1071 | 19.1 |
| 2025-02-01 | Maharashtra | 75.5 | 5.4 | 903 | 11.9 |

Rows: 48 | Columns: 6 | Dimensions: Month, Region | Measures: Sales, Profit, Customers, Discount_pct

Month parsed as Date; numeric measures checked. Data covers Jan–Dec 2025 × 4 regions.

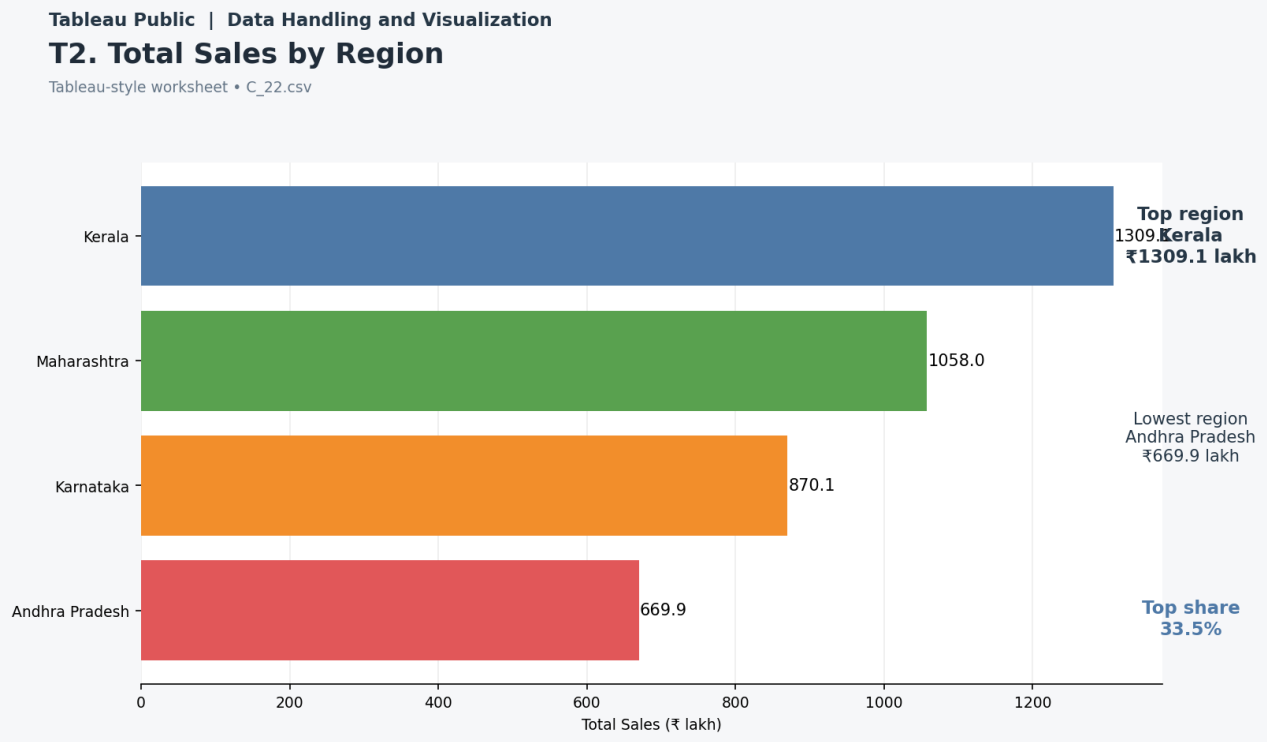
## T2. Bar Chart — Sales by Region

Build a horizontal bar chart with Region and SUM(Sales), sort descending, and turn on mark labels.

Highest sales region: Kerala — ₹1309.1 lakh.

Lowest sales region: Andhra Pradesh — ₹669.9 lakh.

The top region contributes 33.51% of total regional sales.



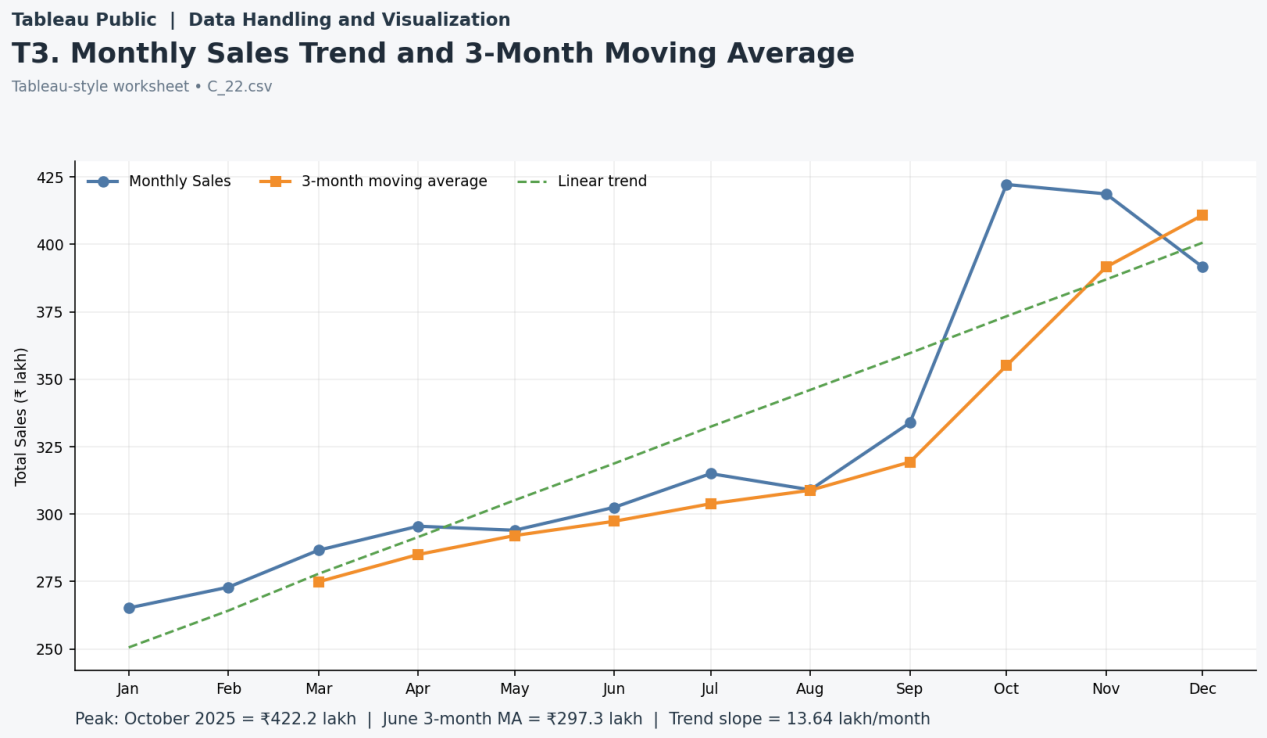

# T3. Line Chart and Trend

The line chart displays total monthly sales. The orange line is the 3-month moving average and the dashed green line is the fitted linear trend.

Peak month: October 2025, with total sales of ₹422.2 lakh.

June 3-month moving average: ₹297.33 lakh (average of April, May, and June totals).

Linear trend slope: 13.64 lakh per month. A positive slope indicates an overall upward trend; a negative slope indicates a downward trend.



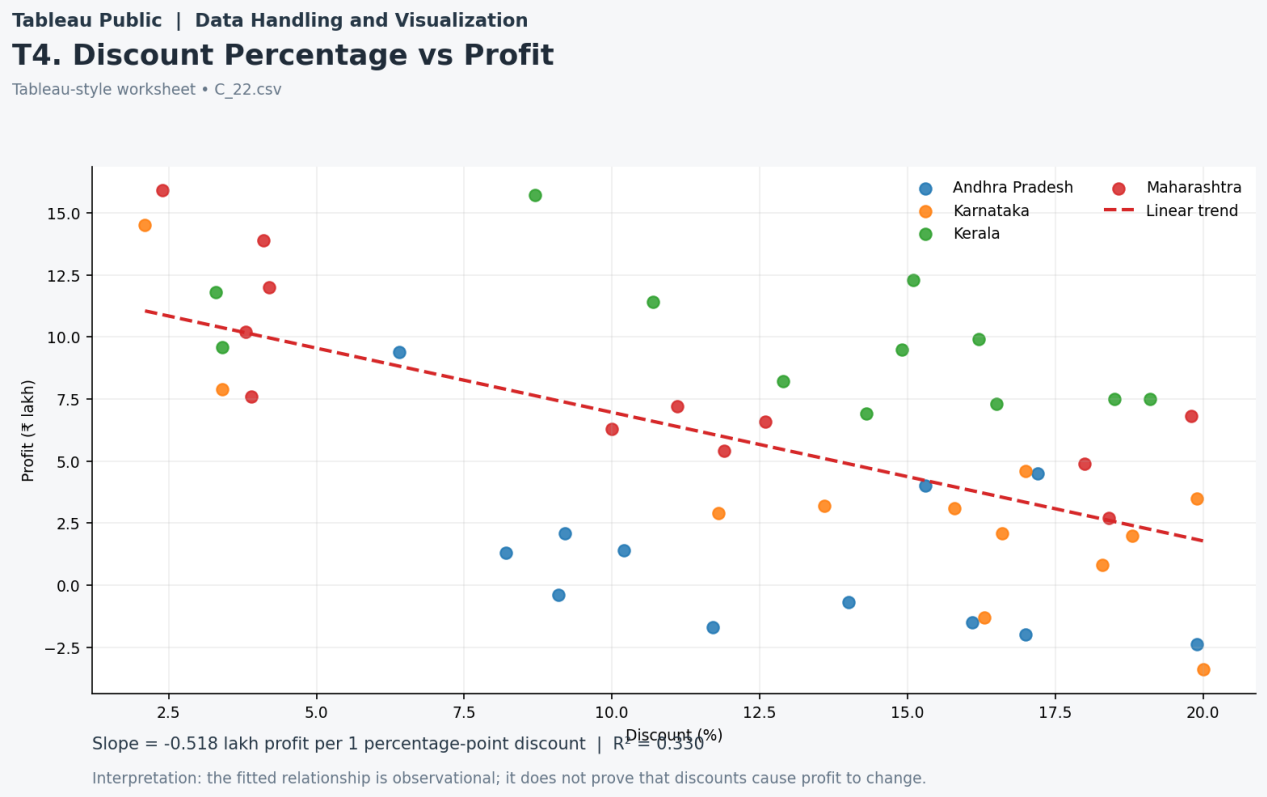

# T4. Scatterplot — Discount vs Profit

Use Discount_pct on Columns and Profit on Rows, keep one mark per record, color by Region, and add a linear trendline.

Fitted slope: -0.518 ₹ lakh of profit per one percentage-point increase in discount.

$R^2$: 0.330. This indicates the fraction of variation in Profit explained by the fitted linear relationship with Discount_pct in this simple model.

Interpretation: the relationship in this dataset is observational and should not be treated as proof that changing discounts alone causes profit to rise or fall.



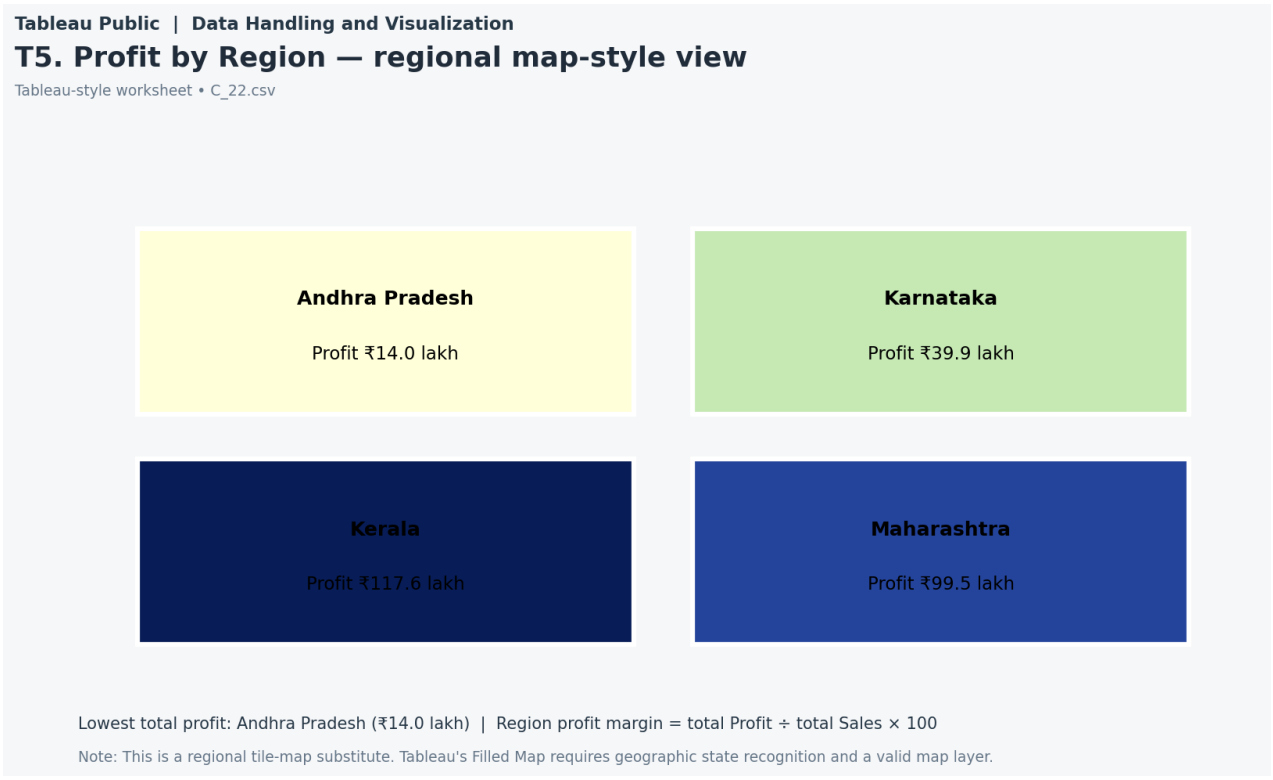

# T5. Filled Map — Profit by Region

In Tableau, double-click Region and use SUM(Profit) as the color measure. If Tableau does not recognize a region, assign the geographic role as State/Province and confirm the country context.

Lowest total profit region: Andhra Pradesh — ₹14.0 lakh.

Profit margin is calculated as total Profit ÷ total Sales × 100. Lowest-profit region margin: 2.09%.

The visual is a tile-map substitute, not a geographic map screenshot. A true Tableau filled map depends on successful geographic matching.



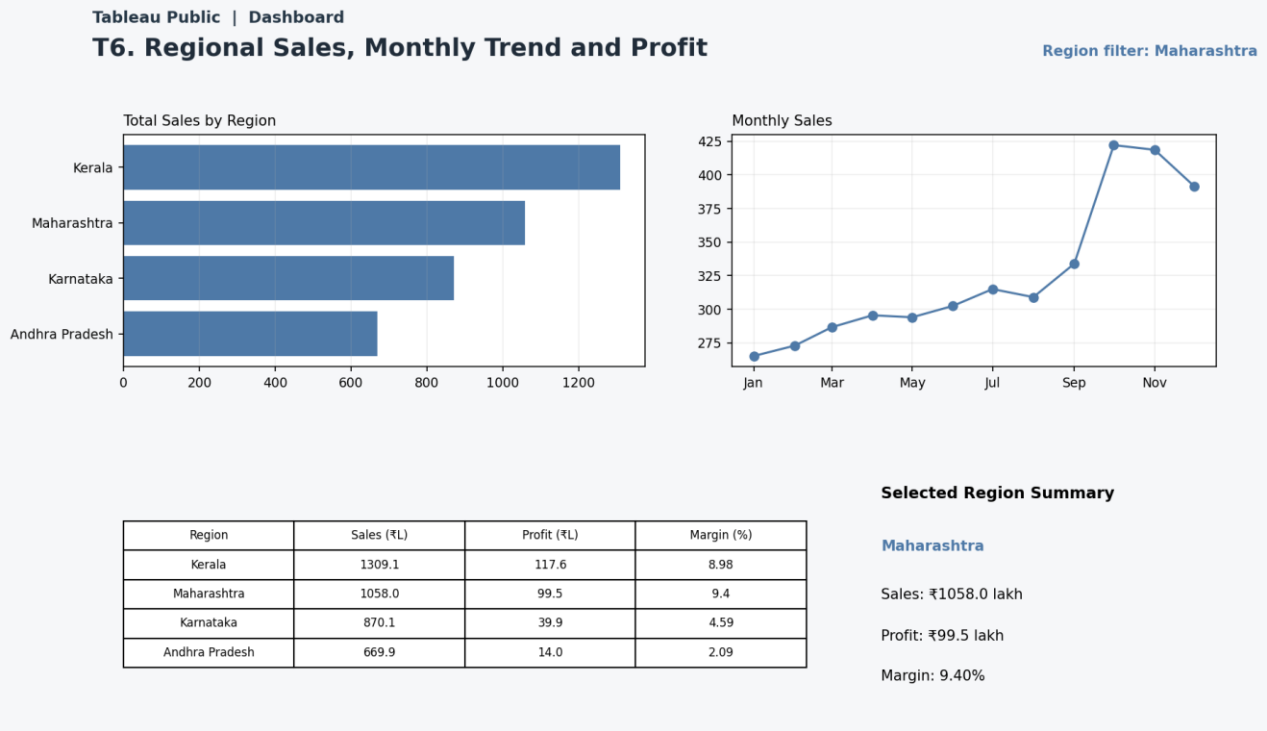

## T6. Dashboard with Region Filter

Place the regional bar chart, monthly sales line chart, and profit map/tile-map on one dashboard. Show the Region filter and apply it to all worksheets.

Second-highest sales region selected: Maharashtra.

Selected-region total sales: ₹1058.0 lakh; total profit: ₹99.5 lakh; profit margin: 9.40%.

To reproduce the interaction in Tableau, select the second-highest region from the filter control and ensure the filter applies to all related worksheets.




| Region | Sales (₹L) | Profit (₹L) | Margin (%) |
|---|---|---|---|
| Kerala | 1309.1 | 117.6 | 8.98 |
| Maharashtra | 1058.0 | 99.5 | 9.4 |
| Karnataka | 870.1 | 39.9 | 4.59 |
| Andhra Pradesh | 669.9 | 14.0 | 2.09 |

## T7. Compare Tableau with R (ggplot2)

Tableau is mainly a drag-and-drop visual analytics tool, useful for rapid exploration, filters, and interactive dashboards. R ggplot2 is code-based and useful for reproducible analysis, customization, and automated reports.

Both tools can create bar charts, time-series line charts, scatterplots, trend lines, and regional summaries. The calculations should match when the same data, aggregation, and statistical method are used.

The image provides a concise comparison table rather than a dataset chart because this task asks for two real differences between the tools.

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## T7. Tableau vs R (ggplot2)

Tableau-style worksheet • C_22.csv

| Aspect | Tableau | R ggplot2 |
|---|---|---|
| Workflow | Drag-and-drop; interactive worksheets | Code-based; reproducible scripts |
| Charts | Fast dashboards, filters, maps | Highly customizable static/interactive plots |
| Trendline / moving average | Analytics pane and quick table calculations | geom_smooth(), rollmean()/rolling functions |
| Dashboard sharing | Packaged workbook / Tableau Public link | R Markdown, Shiny, Quarto or exported files |
| Strength | Easy exploration and presentation | Flexible analysis and automation |

Both tools can produce equivalent analytical findings; Tableau emphasizes interactive exploration, while R emphasizes script-based reproducibility.

# Summary of Computed Regional Results

| Region | Sales (₹ lakh) | Profit (₹ lakh) | Customers | Profit Margin (%) |
|---|---|---|---|---|
| Andhra Pradesh | 669.9 | 14.0 | 7053 | 2.09 |
| Karnataka | 870.1 | 39.9 | 10755 | 4.59 |
| Kerala | 1309.1 | 117.6 | 14756 | 8.98 |
| Maharashtra | 1058.0 | 99.5 | 12761 | 9.40 |

All numerical answers are calculated from the supplied C_22.csv file. Use Tableau's own worksheets to reproduce the interaction and map features for a genuine application screenshot.